Negative Viscosity in Magnetic Fluid (2D Derivation)

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Torque Balance

$$T_{\text{visc}} = -4\pi\eta a^2(\omega_p - \Omega) \tag{1}$$

Symbols:

- T_{visc}: Viscous torque
- η: Fluid viscosity (Pa·s)
- a: Particle radius (m)
- ω_p: Particle angular velocity
- Ω: Fluid vorticity ($\frac{1}{2}(\nabla \times \mathbf{v})_z$)

$$\Gamma_{\text{visc}} = -4\eta\phi(\omega_p - \Omega) \tag{2}$$

Symbols:

- Γ_{visc}: Torque density
- ϕ: Area fraction of particles

$$\Gamma_{\text{mag}} = \mathbf{M} \times \mathbf{H} \tag{3}$$

Symbols:

- $\mathbf{M}$: Magnetization
- $\mathbf{H}$: Magnetic field

$$4\eta\phi(\omega_p - \Omega) = \mathbf{M} \times \mathbf{H} \tag{4}$$

Symbols:

- Torque balance condition

Rotational Viscosity

$$\sigma_a = \frac{1}{2}\mathbf{M} \times \mathbf{H} \tag{5}$$

Symbols:

- σ_a: Antisymmetric stress

$$\tau_{\text{total}} = 2\eta\Omega + 2\eta\phi(\Omega - \omega_p) \tag{6}$$

Symbols:

- τ_{total}: Total shear stress

$$\Delta\eta = \eta\phi\left(1 - \frac{\omega_p}{\Omega}\right) \tag{7}$$

Symbols:

- $\Delta\eta$: viscosity

Momentum Equation

$$\rho\frac{d\mathbf{v}}{dt} = -\nabla p + \eta\nabla^2\mathbf{v} + (\mathbf{M}\cdot\nabla)\mathbf{H} + \frac{1}{2}\nabla \times (\mathbf{M} \times \mathbf{H}) \tag{8}$$

Symbols:

- ρ: Fluid density
- $\mathbf{v}$: Velocity
- p: Pressure

Magnetization Dynamics

$$\frac{d\mathbf{M}}{dt} = -\frac{1}{\tau_B}(M - M_0) \tag{9}$$

Symbols:

- τ_B: Brownian relaxation time
- M_0: Equilibrium magnetization

$$\frac{d\mathbf{M}}{dt} = \omega_p \times M - \frac{1}{\tau_B}(M - M_0) \tag{10}$$

$$\omega_p = \Omega + \frac{M \times H}{4\eta\phi} \tag{11}$$

$$\frac{dM}{dt} = \Omega \times M - \frac{1}{\tau_B}(M - M_0) - \frac{1}{4\eta\phi} M \times (M \times H) \tag{12}$$

Fokker-Planck Equation

$$\frac{\partial W}{\partial t} + \frac{\partial}{\partial \theta}(\Omega W) = \frac{1}{2\tau_B}\frac{\partial}{\partial \theta}\left[\frac{\partial W}{\partial \theta} - W\frac{\partial}{\partial \theta}(\xi \cdot \hat{e})\right] \tag{13}$$

Symbols:

- W: Probability distribution
- θ: Angular coordinate
- $\hat{e}$: Orientation unit vector

Magnetization

$$M = \phi M_s L(\xi_e)\frac{\xi_e}{\xi_e} \tag{14}$$

Symbols:

- M_s: Saturation magnetization
- L: Langevin function (2D)

Effective Field Equation

$$\frac{d}{dt}\left(L(\xi_e)\frac{\xi_e}{\xi_e}\right) = \Omega \times \left(L(\xi_e)\frac{\xi_e}{\xi_e}\right) - \frac{1}{\tau_B}L(\xi_e)\frac{(\xi_e - \xi)}{\xi_e} - \frac{1}{2\tau_B\xi_e^2}\left(1 - \frac{2L(\xi_e)}{\xi_e}\right)\xi_e \times (\xi_e \times \xi) \tag{15}$$

Perturbation Solution

$$\tau_B \frac{d\xi_e}{dt} = -\left(\frac{dL}{d\xi_e}\right)^{-1} \frac{L(\xi_e)}{\xi_e}(\xi_e - \xi) \tag{16}$$

$$M^{(1)} = M^{(0)} F(\xi_e) \tau_B (\Omega \times \hat{h}) \tag{17}$$

$$\tau_B \frac{dF}{dt} = 1 - \frac{1}{2}\left(\frac{1}{L(\xi_e)} - \frac{1}{\xi_e}\right) \xi_0 F \cos(\omega t) \tag{18}$$

Torque and Viscosity Parameter

$$\Gamma = -2\eta\phi\Omega[\xi_0 L(\xi_e) F \cos(\omega t)] \tag{19}$$

$$g = \frac{1}{2}\langle \xi_0 L(\xi_e) F \cos(\omega t) \rangle \tag{20}$$

Final Flow Equation

$$\frac{1}{2}\nabla \times (M \times H) = 2\eta\phi g(\nabla \times \Omega) \tag{21}$$

$$\text{Total Force} = -2\eta(1 + \phi g)(\nabla \times \Omega) \tag{22}$$

Final Results

$$\Delta\eta = \eta\phi g \tag{23}$$

$$\omega_p = (1 - g)\Omega \tag{24}$$